Shinnosuke Ono

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Last Updated: Sep. 2026

EDUCATION

Master of Information Science and Technology, University of Tokyo, Japan Apr. 2025 – Mar. 2027 (Expected)
Supervisor: Prof. Masashi Sugiyama
Thesis (in progress): *Mitigating Reward Hacking in Reinforcement Learning from Human Feedback*

Visiting Research Student, Nanyang Technological University, Singapore Jul. 2026 – Dec. 2026
Supervisor: Prof. Bo An

Bachelor of Information Science, University of Tokyo, Tokyo, Japan
Department of Information Science, Faculty of Science Sep. 2022 – Mar. 2025
Supervisors: Prof. Masashi Sugiyama, Prof. Takashi Ishida
Thesis: *Improving Multimodal Contrastive Learning with Positive-Unlabeled Learning*
College of Arts and Sciences Apr. 2021 – Aug. 2022

PUBLICATIONS

Refereed Publications

1. **A Japanese Language Model and Three New Evaluation Benchmarks for Pharmaceutical NLP.** IJCNLP-AACL 2025
S. Ono*, I. Sukeda*, T. Fujii, K. Buma, S. Sasaki. https://aclanthology.org/2025.ijcnlp-long.72/.

Under Review

1. **Mitigating Reward Hacking in RLHF via Advantage Sign Robustness.**
S. Ono, J. Ackermann, S. Nishimori, T. Ishida, M. Sugiyama. https://arxiv.org/abs/2604.02986. Also appears at 2nd Workshop on Epistemic Intelligence in Machine Learning (EIML), ICML 2026.

Preprints

1. **AIRoA MoMa Dataset: A Large-Scale Hierarchical Dataset for Mobile Manipulation.**
R. Takanami, P. Khrapchenkov, S. Morikuni, J. Arima, Y. Takaba, S. Maeda, T. Okubo, G. Sano, S. Sekioka, A. Kadoya, M. Kambara, N. Nishiura, H. Suzuki, T. Yoshimoto, K. Sakamoto, **S. Ono**, H. Yang, D. Yashima, A. Horo, T. Motoda, R. Kawada, H. Ito, K. Fukuda, A. Goto, K. Morinaga, Y. Ikeda, R. Kawada, M. Yoshikawa, N. Kosuge, Y. Noguchi, K. Ota, T. Matsushima, Y. Iwasawa, Y. Matsuo, T. Ogata. https://arxiv.org/abs/2509.25032.

SCHOLARSHIPS

2. **OverSeas PreDoctoral Fellow (OSPDF), Toyota Physical and Chemical Research Institute (Toyota Riken).** 2027 – 2029
Awarded to exceptional Japanese students pursuing a Ph.D. abroad. Provides US$62,000 funding per year for two years covering tuition, insurance, and living costs, and up to three years of additional funding support.

1. **Heiwa Nakajima Foundation Scholarship.** Jul. 2026 – Dec. 2026
Awarded to outstanding Japanese graduate students studying abroad. Covered my stay at NTU Singapore.

RESEARCH AND WORK EXPERIENCE

Research Assistant, Matsuo-Iwasawa Laboratory, The University of Tokyo Mar. 2025 – Present
Under Prof. Tatsuya Matsushima

- Contributed to the AIRoA MoMa dataset, a large-scale hierarchical dataset for mobile manipulation
- Served as an organizing member of the Physical AI Robot Challenge 2026, a national competition on robot control held jointly with AIRoA. Proposed the benchmark for compositional generalization of vision-language-action models used throughout the contest

Undergraduate Research, The University of Tokyo Sep. 2024 – Mar. 2025

- Improved multimodal contrastive learning using positive-unlabeled learning, under Prof. Masashi Sugiyama and Prof. Takashi Ishida

Lead Engineer, EQUES Inc. Mar. 2024 – Present

- Led the development of JPharmatron, a Japanese language model for the pharmaceutical domain, and JPharmaBench, three evaluation benchmarks for the same domain
- Led the development and evaluation of a vision-language-action model for laboratory automation, selected for the Physical AI Development Support Program by AWS Japan

AWARDS

1. **Rank A, 29th Japan Mathematical Olympiad.** 2019
   Top 7% of 4,423 participants.

TALKS / PRESENTATIONS

2. **Mitigating Reward Hacking in RLHF via Advantage Sign Robustness.** Apr 9, 2026
   Poster presented at Statistical Safeguarding Workshop 2026.

1. **JPharmatron & JPharmaBench: Japanese LLM and Evaluation Benchmarks for Pharmaceutical NLP.** Jan 13, 2026
   The 25-th LLM-jp seminar.

TEACHING

1. **Teaching Assistant, Continuous Algorithm, The University of Tokyo** Oct. 2025 – Mar. 2026
   Course taught by Prof. Hayata Yamasaki. Delivered four lectures and wrote their assignments; graded student work across all sessions.

ACADEMIC SERVICES

1. **Session Chair, Statistical Safeguarding Workshop 2026** Apr 9, 2026
   Hosted by The University of Tokyo & RIKEN AIP Imperfect Information Learning Team.
   https://statsafe.github.io/workshop20260409.html.

COMMUNITY SERVICES

1. **How does AI memorize knowledge?** Oct. 10, 2025
   Gave a talk at a lab visit from Matsue-Minami High School.

SKILLS

Language Skills:

- Japanese: Native
- English: Equivalent to CEFR C1. TOEFL iBT: 107 (2025).
- Spanish: Equivalent to CEFR B2. Completed Trilingual Program (Spanish) of the University of Tokyo in 2022.